

	(Plate Q is grounded, hence is at 0 V.)	
6(a)(ii)	To move undeflected, the net force must be zero. $F_E = F_B$ $qE = qvB$ $v = \frac{E}{B}$	M1 A0
6(a)(iii)	 $v > \frac{E}{B} \Rightarrow qvB > qE$, or $F_B > F_E$. Since the magnetic force F_B points downwards, <u>the particles will be deflected downwards.</u> (Note that the path is not parabolic.)	B1 B1
6(b)(i)	The <u>magnetic force is always perpendicular to the velocity.</u> Hence, <u>it does no work on the particles.</u> The kinetic energy, hence the speed, of the particle remain constant.	B1 B1
6(b)(ii)	The magnetic force provides the centripetal force. $qvB = m \frac{v^2}{r} \Rightarrow \frac{q}{m} = \frac{v}{Br}$ With $v = \frac{E}{B}$, $\frac{q}{m} = \frac{E}{B^2 r}$	M1 M1 A0

Qns	Answer	Marks
3(e)	$t = 1, 2, 3$	